# 2026 Ecosystem Summary Report pilot for Region 4X

Maritimes EBFM WG

2026-03-02

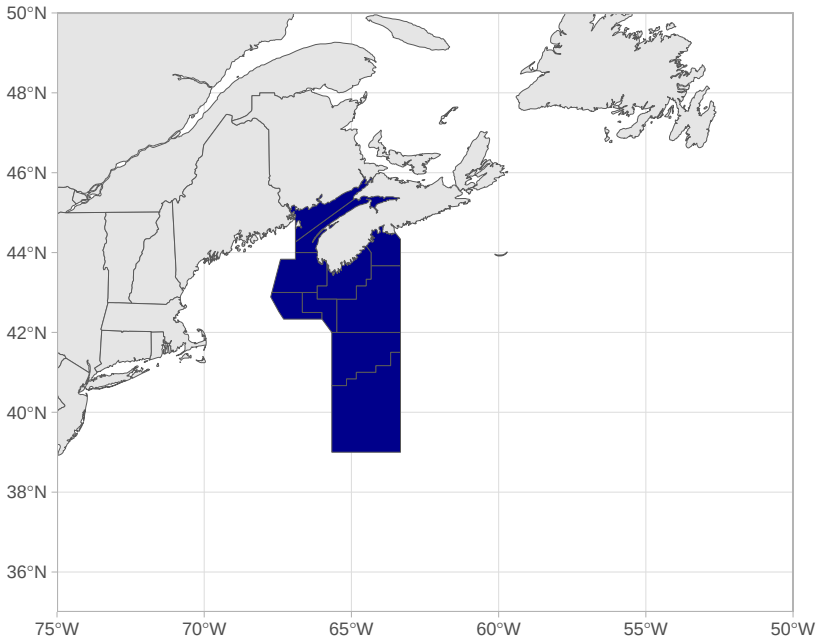


Figure 1: Map of Region: 4X

# Introduction

### About this report

This Ecosystem Summary Report supports DFO's efforts to incorporate ecosystem approaches into Fisheries Science and Resource Management in the Maritimes Region. The purpose of this report is to synthesize ecosystem information to inform an ecosystem approach, as highlighted in the modernized Fisheries Act (2019; Bill C-68). It aligns objectives broadly to those of the Maritimes Ecosystem Based Management Framework, and regional/area-based objectives that contribute to Scientific Advice and Management Plans or other Harvest Strategies. The aim of this Ecosystem Summary Report is to provide an integrative suite of ecosystem indicators relevant to a specific fisheries management area that can support Ecosystem Approaches to Fisheries Management (EAFM) and Ecosystem Based Fisheries Management (EBFM) at DFO.

### Report structure

This report outlines ecosystem objectives related to a specific management area (Table 1), and reviews relevant ecosystem indicator trends and the status of the most recent year(s) relative to a reference point (if available) or a long-term average. The results are synthesized to outline potential implications for the management area examined, but not specifically related to a focal stock or species. Where relevant this document provides links to reports and resources for details and methodology. This report summarizes general ecosystem trends to support integration of ecosystem information and expertise into Science Advice (e.g. stock assessment) or Resource Management (e.g. Integrated Fisheries Management Plans), but is intended for a variety of users.

This is a summary of the best available information describing the ecosystem specific to management areas within the Maritimes Region.

## Ecosystem Objectives

The Ecosystem objectives outline the ecosystem context and considerations for this management area.

Table 1: Ecosystem objectives related to EBM Framework Pillar, Management Components, and Indicators.

| Pillar | Management Component | Ecosystem Objective | Indicators |
|---|---|---|---|
| A. Pressures and stressors | Climate and Oceanography | Monitor key signals within the environment and ecosystem | AMO,NAO, SST anomaly, Bottom Temp anomaly, SST, Bottom Temp, Stratification, Salinity, Ocean Acidification |
| A. Pressures and stressors | Longer-term Climate Change | Effective action and adaptation on climate change is taken | Future projection model outputs |
| B. Ecological | Commercial Fishery | Maintain fishing pressure consistent with the ecosystem's ability to recover | Fishing pressure, Commercial landings |
| B. Ecological | Habitat | Habitat and habitat features are conserved, protected, maintained, and restored | Cumulative impacts |
| B. Ecological | Productivity | Do not cause unacceptable reduction to productivity | Nitrate, Chlorophyll, Bloom statistic, C.finmarchicus, Zooplankton biomass |
| B. Ecological | Productivity | Ecosystem structure and functioning is promoted and conserved | Large Fish Indicator |
| B. Ecological | Biodiversity | Biodiversity is conserved, maintained, and restored to preserve the structure and natural resilience of aquatic ecosystems | Shannon Diversity Index, Margalef Richness, Heips Evenness, Guild-level biomass |
| B. Ecological | Non-Target species | Unintended or incidental mortality for all species is controlled | NA for this report |
| C. Economic | Commercial Fishery | Aquatic resources are managed to foster long term, viable, prosperous and sustainable livelihoods for all users | Landings values |
| D. Social-Cultural | NA for this report | NA for this report | NA for this report |
| E. Governance | NA for this report | NA for this report | NA for this report |

# A. Pressures and Stressors

## Climate and Oceanography

The management area is strongly influenced by natural variability of the climate system (e.g., Atlantic Multidecadal Oscillation, North Atlantic Oscillation) and, therefore, long-term monitoring is needed to determine how anthropogenic climate change is affecting ocean temperatures and other ocean properties. Changes in climate can directly and indirectly impact stock status by affecting the distribution, seasonal timing, physiology, migration and spawning of marine species.

### Atlantic Multidecadal Oscillation (AMO); global scale

The AMO describes long-duration changes in the sea surface temperature of the North Atlantic Ocean, with cool and warm phases that may last for 20-40 years.

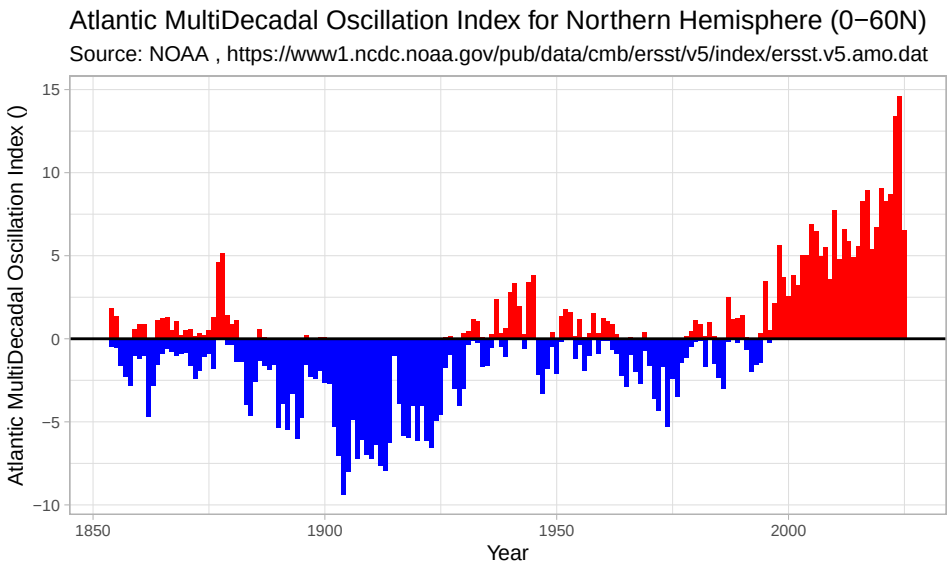


Figure 2: The Atlantic Multidecadal Oscillation (AMO) index

**North Atlantic Oscillation (NAO); global scale**

Hebert et al. 2024 (see Section 2.1)

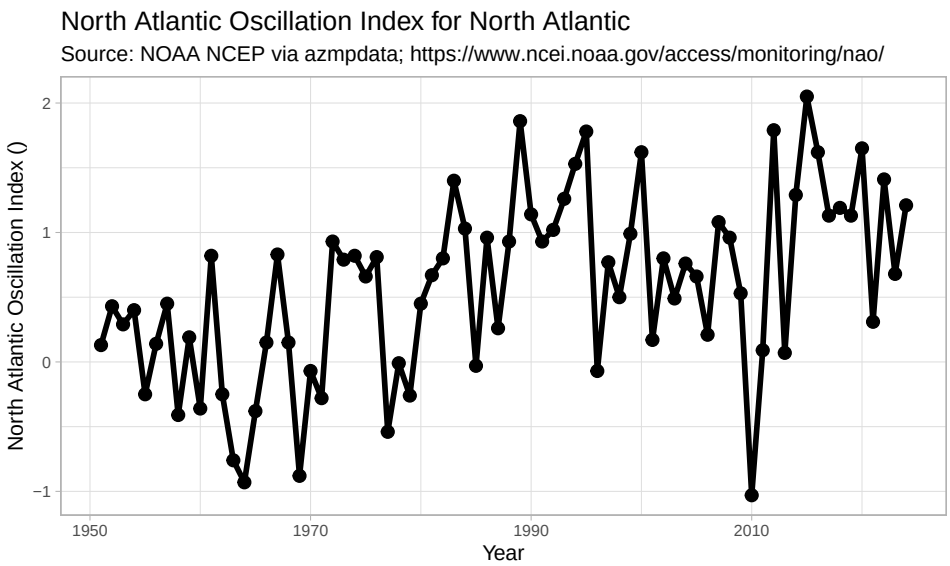


Figure 3: The North Atlantic Oscillation (NAO) index, defined as the winter (December, January, February, March) 500 mb pressure Principal Component Analysis which is representative of the difference between the Icelandic low and Azores high.

**AZMP Surface Temperature from Satellite Data; local scale 4XSS**

Casault et al. 2020

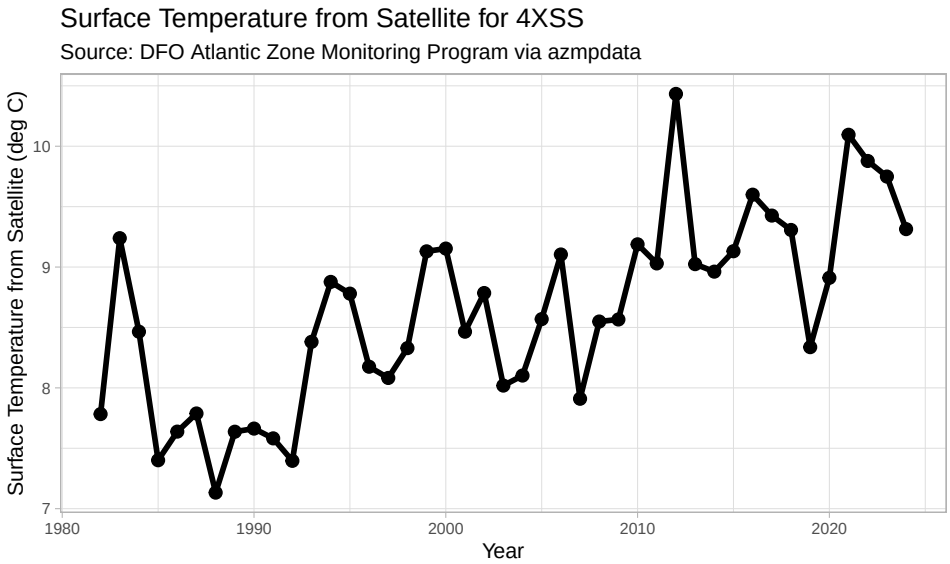


Figure 4: Sea surface temperature from satellite data for region 4X

**AZMP Surface Temperature; local scale (Scotian Shelf Box)**

Casault et al. 2020

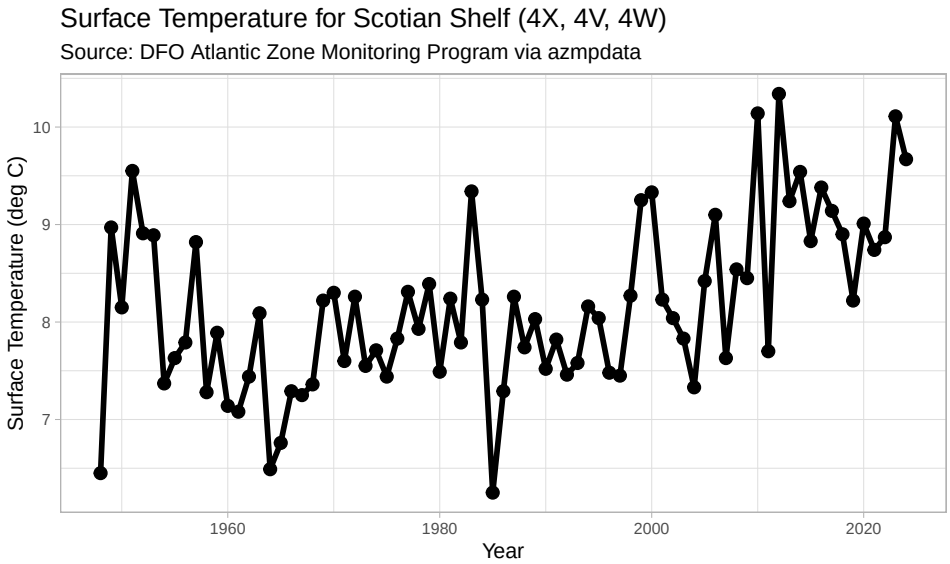


Figure 5: Surface temperature in the Scotian Shelf Region.

**AZMP Sea Bottom Temperature; local scale 4X**

Casault et al. 2020

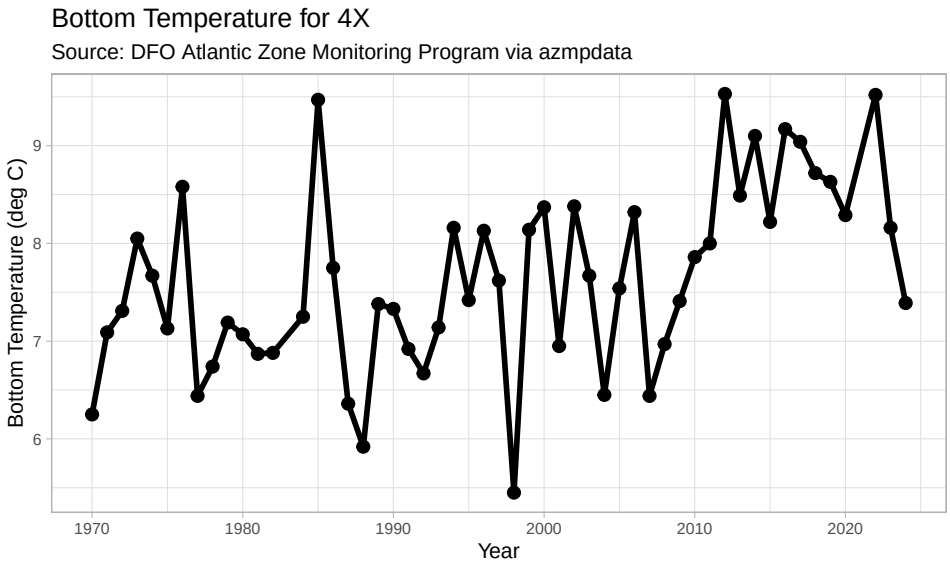


Figure 6: Sea bottom temperature for region 4X

**AZMP Stratification; local scale (Scotian Shelf Box)**

Layton et al. 2025

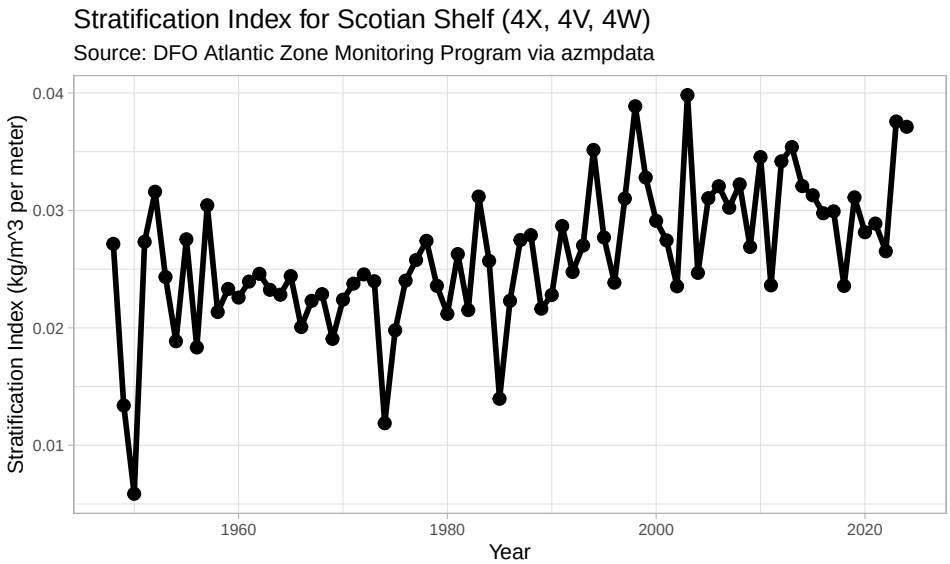


Figure 7: Stratification in the Scotian Shelf Region.

**AZMP Salinity; local scale (Scotian Shelf Box)**

Layton et al. 2025

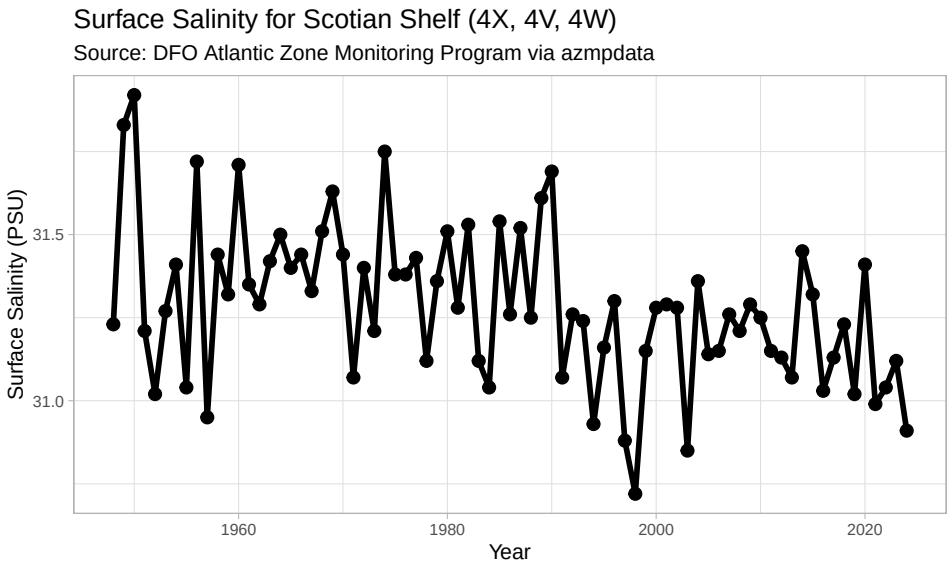


Figure 8: Salinity in the Scotian Shelf Region.

**Ocean Acifification**

Template data not available.

## Longer-Term Climate Change

### Future Projections and Model Outputs

Template data not available.

# B. Ecological

## Commercial Fisheries

### Fisheries Landings; local scale c("4X")

Commercial fishing is an activity that impacts the ecology of a management area through removals of fish from the ecosystem

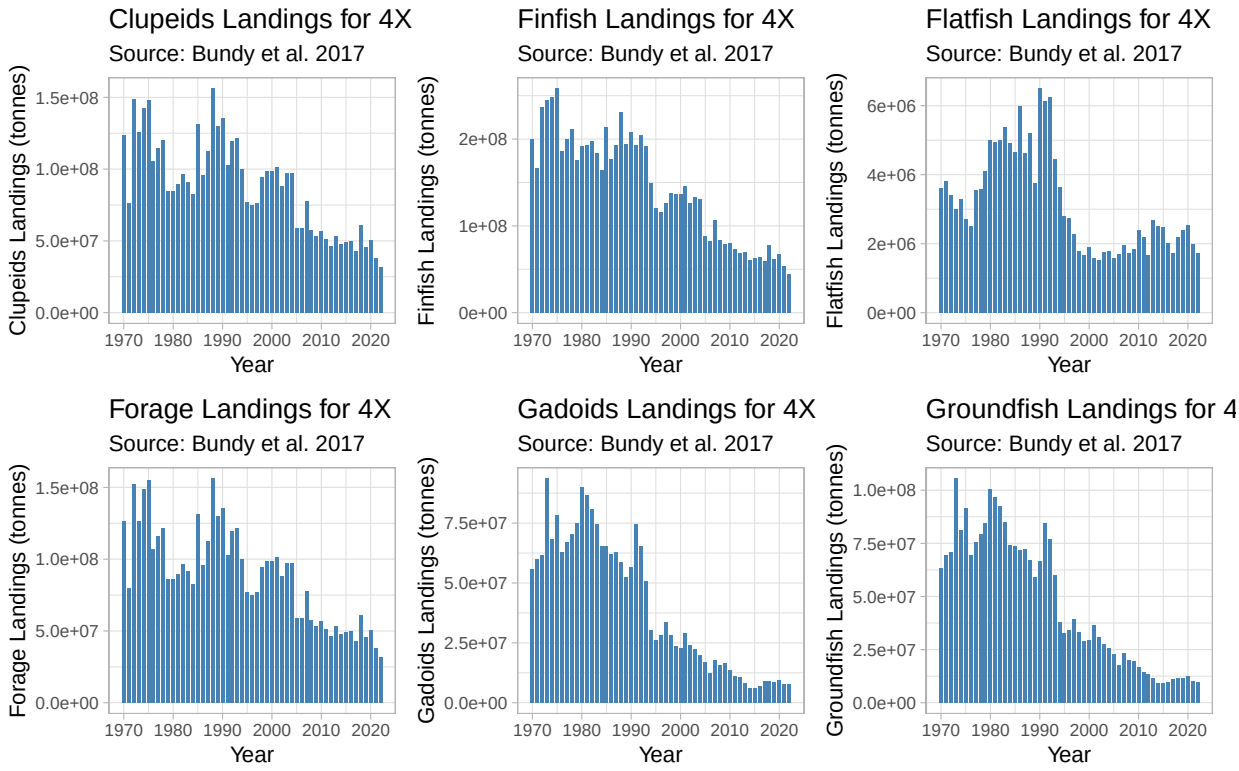


Figure 9: Commercial fisheries landings for target groups in region 4X

**Fisheries Pressure; local scale c("4X")**

This indicator measures the level of exploitation or total fishing pressure at the ecosystem level (Landings/Biomass). Change in this indicator can result from change in biomass, landings or both.

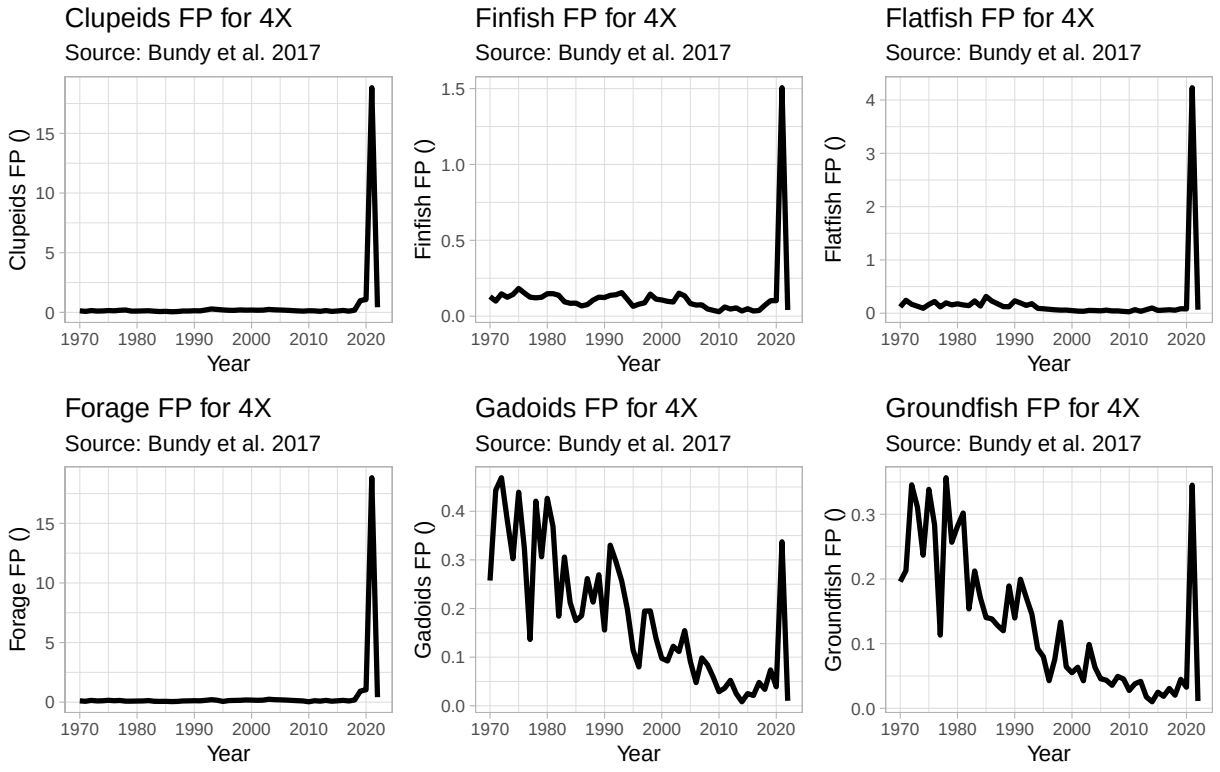


Figure 10: Fishing pressure for target groups in region 4X

## Habitat

### Cumulative Impacts

Template data not available.

## Productivity

### Nitrate

Template data not available.

### Bloom Statistics

Template data not available.

### Zooplankton Biomass

Template data not available.

### Large Fish Indicator

Template data not available.

## Biodiversity

### Shannon Diversity Index, Margalef Richness, Heip's Evenness; local scale c("4X")

Shannon Diversity Index accounts for both abundance and evenness of species present in a community. Margalef Richness measures the number of species present accounting for sampling effects. Heips Evenness measures how equally the species richness contributes to the total abundance or biomass of the community. Due to improvements in species identification over time, long-term Shannon Diversity and Margalef richness can be misleading, only years 2000-2020 were used here.

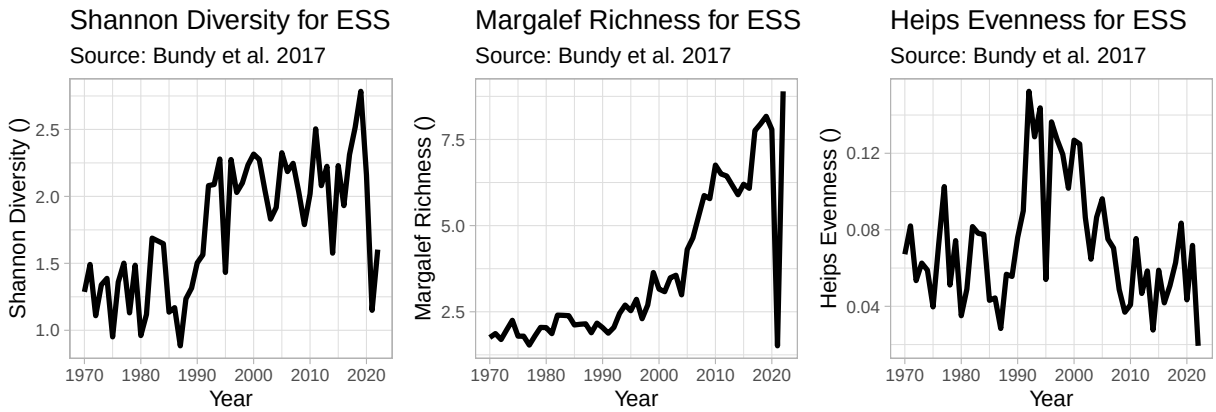


Figure 11: Biodiversity Variables for target groups in region 4X

**Guild-Level Biomass; local scale c("4X")**

Guild-level biomasses address structural attributes of food webs, and can also serve as proxies for ecosystem functioning (Tam et al. 2017). This indicator includes multiple guilds for fish and invertebrates from the RV Survey.

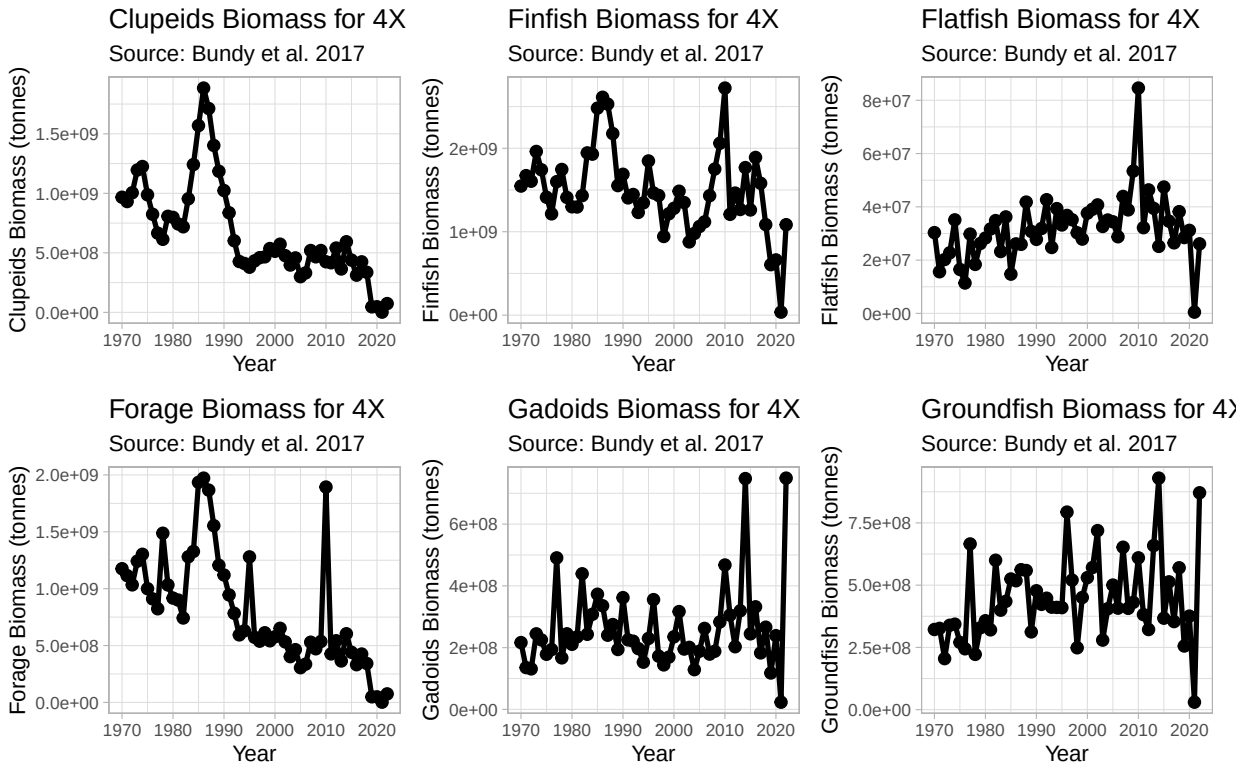


Figure 12: Biomass of target groups from RV surveys in region 4X

## Non-Target Species

### Non-Target Species

Template data not available.

# C. Economic

## Commercial Fisheries

Commercial fishing is an important activity in the Maritimes Region that provides both cultural and economic value to coastal communities.

### Landing Value by Species Category

Template data not available.

## D. Social-Cultural

Currently unavailable

## E. Governance

Currently unavailable